

Hypothetical Analysis and Questions

Sam Harper

Overview

The code below attempts to create a simulated dataset for the purposes of evaluating how factors such as the level of expertise (high, medium, low, not enough) and role (reviewer, panelist) might impact overall scores. Ultimately we are interested in how these factors might affect the difference between initial consensus and final overall scores and, ultimately, funding decisions, but that depends largely on what data are available.

The unit of observation is the application and basic structure attempts to mimic the multilevel nature of the review process for the CIHR Project Grant competitions. There are roughly 50 CIHR Committees, and each committee has around 25 or so members.¹ In practice the number of total applications may vary quite a lot across committees (e.g., up to 50 for PH1 or PH2 when I was SO); however since we are focused on the evaluation of overall scores *among those proposals discussed* (i.e., excluding those streamlined), this is likely closer to 15 or so proposals per committee (again, my reference is PH committees).

In the code below we create some fake data on committees, discussed applications, and members, as well as some simple coefficients for reviewer status and expertise (i.e., with no interaction between these factors).

First, load the packages needed for this setup and analysis

```
# list of packages needed
pkgs <- c('here', 'tidyverse', 'faux', 'modelsummary',
          'fixest', 'tinytable', 'marginaleffects',
          'truncnorm', 'lme4')

# install any needed packages
# install.packages(pkgs)

# load all packages at once
lapply(pkgs, library, character.only=TRUE)
```

Now we set up simple parameters and multilevel structure for the simulated data. For simplicity we take 50 committees and 24 members per committee, though in practice it seems likely that this would vary across committees. Note that since we are focused on those applications that are not streamlined and make it to discussion, we limit the number of applications to 15 per committee (again, this would probably vary across committees). In addition, given this is restricted to the

¹For the Project Scheme in 2024 the number of members per committee was 31.2 (SD=1.9) from a low of about 28 to a high of about 34

discussion phase the distribution of the overall score will be truncated. CIHR places an upper limit of 4.9 for the highest ranking and in many cases a lower level of 3.5 is used (though not the only criteria) to draw the line below which applications are streamlined.

The baseline score here is set to 4.1 with random variation across committee, application, and committee member, as well as are small effects being a panelist (small positive rating) and different levels of expertise (high, medium, low, not enough).

```
# set seed for reproducibility
set.seed(4875)

# define parameters
cmte_n = 50      # number of committees
app_n = 15       # number of discussed applications
mem_n = 24       # number of committee members
b0 = 4.1         # intercept for average score
b1 = 0.2         # fixed effect of panelist vs. reviewer
b2 = -0.1        # fixed effect of high expertise
b3 = 0.1         # fixed effect of low expertise
b4 = 0.2         # fixed effect of no expertise
u0c_sd = 0.1     # random intercept SD for committee
u0m_sd = 0.2     # random intercept SD for members
u0a_sd = 0.2     # random intercept SD for applications
sigma_sd = 0.2   # error SD for overall scores
score_min = 3.5  # lower bound for score
score_max = 4.9  # upper bound for score

# set up data structure
data <- add_random(committee = cmte_n,
  application = app_n, member = mem_n) |>

# recode values for committee, application, and member
add_between("committee",
  cmte = sprintf("%02d", 1:cmte_n)) |>
add_between("application",
  app = 1:app_n) |>
add_between("member",
  memno = sprintf("%02d", 1:mem_n)) |>

# create unique ID for each committee member
mutate(cid = paste0(cmte, "_", memno)) |>

# assign reviewers uniquely within each application
group_by(cmte, app) |>
mutate(
  job = sample(c(rep("reviewer", 3),
    rep("panelist", 21))),
  # add expertise for each member
  exp = sample(c(rep("high", 6),
    rep("med", 10), rep("low", 4),
    rep("none", 4)))) |>
```

```

ungroup() |>

# add indicators for reviewer, expertise
mutate(
  panelist = if_else(job == "panelist", 1, 0),
  exp_high = if_else(exp == "high", 1, 0),
  exp_low = if_else(exp == "low", 1, 0),
  exp_none = if_else(exp == "none", 1, 0)
) |>

# add random effects
add_ranef("cmte", u0c = u0c_sd) |>
add_ranef("member", u0m = u0m_sd) |>
add_ranef("application", u0a = u0a_sd) |>
add_ranef(sigma = sigma_sd) |>

# Compute score using a truncated normal distribution
mutate(
  score = rtruncnorm(n(), a = score_min, b = score_max,
    mean = b0 + u0c + u0m + u0a +
      (b1 * panelist) + (b2 * exp_high) +
      (b3 * exp_low) + (b4 * exp_none),
    sd = sigma_sd)
) |>

# drop intermediate variables
select(-committee, -application, -member,
  -u0c, -u0m, -u0a, -sigma)

# write a copy as a csv file
write_csv(data, here("data", "sim-data.csv"))

```

Here is a glimpse of the data structure:

```

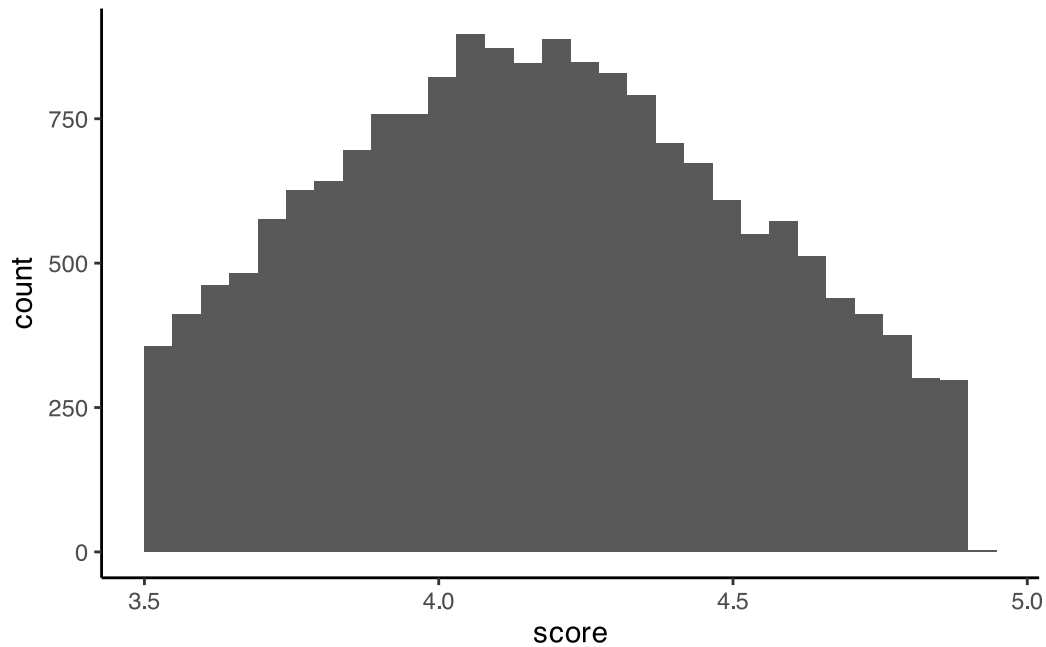
tt(head(data), digits = 2) |>
style_tt(fontsize = 0.8)

```

cmte	app	memno	cid	job	exp	panelist	exp_high	exp_low	exp_none	score
01	1	01	01_01	panelist	none	1	0	0	1	4.4
01	1	02	01_02	panelist	med	1	0	0	0	4.5
01	1	03	01_03	panelist	med	1	0	0	0	3.9
01	1	04	01_04	reviewer	none	0	0	0	1	3.9
01	1	05	01_05	panelist	high	1	1	0	0	4
01	1	06	01_06	panelist	low	1	0	1	0	4.4

And a simple histogram of the distribution of overall scores:

```
ggplot(data, aes(x = score)) + geom_histogram() +
  theme_classic()
```



A simple set of models with random effects for committee, member, and application, and fixed effects for whether or not the score comes from a reviewer or panelist and various levels of expertise:

```
# empty
m0 <- lmer(score ~ 1 + (1 | cmte) + (1 | memno) +
  (1 | app), data = data)

# add reviewer
m1 <- lmer(score ~ 1 + factor(job) + (1 | cmte) +
  (1 | memno) + (1 | app), data = data)

# add expertise
m2 <- lmer(score ~ 1 + factor(job) + factor(exp) +
  (1 | cmte) + (1 | memno) + (1 | app),
  data = data)

modelsummary(list("Empty" = m0,
  "+ Reviewer" = m1, "+ Expertise" = m2),
  gof_omit = 'DF|Deviance|R2|AIC|BIC|RMSE',
  escape = TRUE)
```

	Empty	+ Reviewer	+ Expertise
(Intercept)	4.172	4.194	4.087

	Empty	+ Reviewer	+ Expertise
	(0.064)	(0.064)	(0.064)
factor(job)reviewer		-0.171 (0.005)	-0.170 (0.004)
factor(exp)low			0.171 (0.004)
factor(exp)med			0.083 (0.004)
factor(exp)none			0.259 (0.004)
SD (Intercept cmte)	0.078	0.078	0.078
SD (Intercept memno)	0.190	0.189	0.188
SD (Intercept app)	0.191	0.191	0.191
SD (Observations)	0.216	0.208	0.189
Num.Obs.	18000	18000	18000
ICC	0.6	0.6	0.7

In the above simulation these effects are simply additive, but it is possible to generate predicted scores for applications that happen to have specific characteristics. For example, we can contrast the predicted scores for a panelist with “Not enough” expertise with that of a reviewer with high expertise:

```
p1 <- predictions(m2,
  newdata = datagrid(job = unique, exp = unique))

p1 |> select(job, exp, estimate, conf.low, conf.high) |>
  tt(digits = 3) |>
  style_tt(fontsize = 1)
```

job	exp	estimate	conf.low	conf.high
panelist	none	4.14	4.02	4.27
panelist	med	3.97	3.84	4.09
panelist	high	3.88	3.76	4.01
panelist	low	4.05	3.93	4.18
reviewer	none	3.97	3.85	4.1
reviewer	med	3.8	3.67	3.92
reviewer	high	3.71	3.59	3.84

job	exp	estimate	conf.low	conf.high
reviewer	low	3.88	3.76	4.01

This toy example leads to average predictions of 4.1 for a panelist with “Not enough” experience vs. 3.7 for a reviewer with a “high” level of expertise. To the extent that committees vary in their composition of expertise, this could have some impact on overall scores.

Potential extensions

In large part the simplified dataset above only allows for basic questions regarding how much, for example, expertise and reviewer status may affect the overall score. More interesting questions could be asked if additional fields are available at different levels of the data.

Application level

Additional data on funding result, initial reviewer scores, consensus score, keywords or domain of inquiry, re-submission status, no. of investigators, funding requested. This would allow for a better disentangling of how, for example, initial reviewer scores and consensus scores may be altered by committee discussion, as well as how this may have affected which applications were ultimately funded.

Reviewer level

Data on reviewer gender, experience, past funding success, conflicts of interest. If there are additional fields that could be (safely) added on reviewer characteristics, this could help further answer questions about how the composition of committees may also impact overall scores and funding.

Applicant level

Lastly, data (respecting confidentiality) on gender, career stage, age, scientific productivity, prior funding success, would allow for the investigation of the impact that committee discussions may have on scores and success rates by important applicant characteristics. While not feasible in a packaged dataset, conducting these analyses remotely would still allow important questions to be answered.

Questions

- Are any of the other fields of data mentioned above available for analysis?
- If so, which of these may be possible to create as a data extract and which might feasibly be analyzed by CIHR analysts in collaboration with our team?
- How might we budget for the amount of time needed to create datasets or run analyses remotely?